Unit 3 Classwork — Equations & Inequalities — Answer Key

One/Two-Step · Distributive · Both Sides · Multi-Step · Special Cases · Inequalities · Word Problems

A. One- & Two-Step Equations

Problem 1 Score 3

Solve: $4x = 28$

Answer: 7

Problem 2 Score 3

Solve: $3x + 5 = 20$

Answer: 5

Problem 3 Score 3

Solve: $\frac{x}{3} - 2 = 4$

Answer: 18

Problem 4 Score 4

Solve: $-2x + 7 = 1$

Answer: 3

B. Distributive Property & Variables on Both Sides

Problem 1 Score 3

Solve: $3(2x - 1) = 15$

Answer: 3

Problem 2 Score 4

Solve: $7x - 2 = 3x + 14$

Answer: 4

C. Multi-Step Equations

Problem 1 Score 4

Solve: $3(x + 2) = 2x + 10$

Answer: 4

Problem 2 Score 4

Solve: $5 - 2x = 3x + 15$

Answer: −2

Problem 3 Score 5

Solve: $4x - 3 = 2(x + 5)$

Answer: 6.5

D. Special Cases — No Solution or Infinite Solutions

Problem 1 Score 4

Solve: $2x + 5 = 2x - 1$ — how many solutions?

Answer: no solution

Problem 2 Score 5

Solve: $3(x + 4) = 3x + 12$ — how many solutions?

Answer: infinitely many solutions

E. Linear Inequalities

Problem 1 Score 3

Solve: $3x \leq 21$

Tap to type: x | < | > | ≤ | ≥ | − | ⌫ Erase

Answer: $x \leq 7$

Problem 2 Score 3

Solve: $5x - 3 \geq 12$

Tap to type: x | < | > | ≤ | ≥ | − | ⌫ Erase

Answer: $x \geq 3$

Problem 3 Score 4

Solve: $-2x > 8$ (careful!)

Tap to type: x | < | > | ≤ | ≥ | − | ⌫ Erase

Answer: $x < -4$

Problem 4 Score 5

Solve: $-3x + 1 \leq 10$ (careful!)

Tap to type: x | < | > | ≤ | ≥ | − | ⌫ Erase

Answer: $x \geq -3$

F. Word Problems

Problem 1 Score 4

A taxi charges $3 plus $2 per mile. If the total fare was $17, how many miles were driven?

Answer: 7

Problem 2 Score 5

A gym membership costs $25 to join plus $15/month. You can spend at most $100. What's the max number of months?

Answer: 5